

JOINS:

"The process of retrieval of data from multiple tables simultaneously is known as JOINS".

WHEN/ WHY do we use joins:

"whenever we have to select attribute from more than one table then we use joins"

Types of JOINS.

1. CARTESIAN JOIN / CROSS JOIN
2. INNER JOIN / EQUI JOIN (instead of subquery case2)
3. OUTER JOIN
 - a) LEFT OUTER JOIN
 - b) RIGHT OUTER JOIN
 - c) FULL OUTER JOIN
4. SELF JOIN (instead of subquery case1 and case3)
5. NATURAL JOIN .

1. CARTESIAN JOIN / CROSS JOIN :

In Cartesian Join a record from table 1 will be merged with All the records of table 2

Number of Columns in the Result table : will be equivalent to the summations of columns present in both the tables .

Number of Col = Number of Col T1 + Number of Col T2

$$2 + 2$$

= 4 Columns

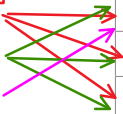
Number of Rows in the Result table : will be equivalent to the product of number of rows present in the both the tables .

Number of Rows = Number of Rows T1 x Number of Rows T2

$$3 \times 3 = 9 \text{ Rows .}$$

Example

<u>EMP</u>		<u>DEPT</u>	
ENAME	DEPTNO	DNAME	DEPTNO
A	30	D1	10
B	10	D2	20
C	20	D3	30



Result Table :

ENAME	DEPTNO	DNAME	DEPTNO
A	30	D1	10
A	30	D2	20
A	30	D3	30
B	10	D1	10
B	10	D2	20
B	10	D3	30
C	20	D1	10
C	20	D2	20
C	20	D3	30

SYNTAX:

1. ANSI [American National Standard Institute]

```
SELECT Column_Name  
FROM Table_Name1 CROSS JOIN Table_Name2 ;
```

2. Oracle

```
SELECT Column_Name  
FROM Table_Name1 , Table_Name2 ;
```

Example :

WAQTD ename and dept name for all the employees .

```
SELECT ENAME,DNAME  
FROM EMP,DEPT;  
OR  
SELECT ENAME,DNAME  
FROM EMP CROSS JOIN DEPT;
```

Drawback-→ the count of invalid records will be more than the count of valid records

INNER JOIN / EQUI JOIN:

"It is used to Obtain only Matching Records " Or " A records which has a Pair " .

EMP		DEPT	
ENAME	DEPTNO	DNAME	DEPTNO
A	20	D1	10
B	30	D2	20
C	10	D3	30

JOIN Condition : It is a condition on which the two tables Are merged .

Syntax: Table_Name1.Col_Name = Table_Name2.Col_Name

Join Condition : **EMP.DEPTNO = DEPT.DEPTNO**

20=10 F	30=10 F	10=10 T
20=20 T	30=20 F	10=20 F
20=30 F	30=30 T	10=30 F

Result Table :

ENAME	DEPTNO	DNAME	DEPTNO
A	20	D2	20
B	30	D3	30
C	10	D1	10

SYNTAX:

1. ANSI [American National Standard Institute]

```
SELECT Column_Name  
FROM Table_Name1 INNER JOIN Table_Name2  
ON < JOIN_CONDITION > ;
```

```
SELECT *  
FROM EMP INNER JOIN DEPT  
ON EMP.DEPTNO = DEPT.DEPTNO ;
```

2. Oracle

```
SELECT Column_Name  
FROM Table_Name1, Table_Name2  
WHERE <JOIN_CONDITION > ;
```

```
SELECT *  
FROM EMP , DEPT  
WHERE EMP.DEPTNO = DEPT.DEPTNO ;
```

1. WAQTD ename and dept name for all the employees .

```
SELECT ENAME,DNAME  
FROM EMP,DEPT  
WHERE EMP.DEPTNO=DEPT.DEPTNO ;
```

2. WAQTD ename and loc for all the employees working as Manager .

```
SELECT ENAME,LOC  
FROM EMP,DEPT  
WHERE EMP.DEPTNO=DEPT.DEPTNO AND JOB='MANAGER';
```

3. WAQTD ename , sal and dname of the employee working as Clerk in dept 20 with a salary of more than 1800 .

```
SELECT ENAME,SAL,DNAME  
FROM EMP, DEPT  
WHERE EMP.DEPTNO =DEPT.DEPTNO AND JOB='CLERK' AND DEPTNO=20 AND SAL>1800;
```

COLUMN IS AMBIGUOUSLY DEFINED

```
SELECT ENAME,SAL,DNAME  
FROM EMP, DEPT WHERE EMP.DEPTNO =DEPT.DEPTNO AND JOB='CLERK' AND  
DEPT.DEPTNO=20 AND SAL>1800;
```

4. WAQTD ename,deptno , dname and loc of the employee earning more than 2000 in New York .

```
SELECT ENAME,DNAME,LOC,DEPTNO  
FROM EMP,DEPT  
WHERE EMP . DEPTNO=DEPT.DEPTNO AND SAL>2000 AND LOC='NEW YARK';
```

```
SELECT ENAME,DNAME,LOC,EMP.DEPTNO  
FROM EMP,DEPT  
WHERE EMP . DEPTNO=DEPT.DEPTNO AND SAL>2000 AND LOC='NEW YARK';
```

OUTER JOIN:

"It is used to Obtain Un-Matched Records "

It will provide valid records from all tables and invalid records for any one of the table or all the table

There are 3 types:

Left Outer join

Right outer join

Full outer join

Left Outer Join:

"It is used to obtain Un-Matched Records of Left Table Along with Matching Records ".

SYNTAX:

ANSI [American National Standard Institute]

```
SELECT Column_Name  
FROM Table_Name1 LEFT | OUTER | JOIN Table_Name2  
ON < JOIN_CONDITION > ;
```

```
SELECT *  
FROM EMP LEFT JOIN DEPT  
ON EMP.DEPTNO = DEPT.DEPTNO ;
```

2. Oracle

```
SELECT Column_Name  
FROM Table_Name1 , Table_Name2  
WHERE Table1.Col_Name = Table2.Col_Name (+) ;
```

```
SELECT *  
FROM EMP , DEPT  
WHERE EMP.DEPTNO = DEPT.DEPTNO (+);
```

OUTER JOIN

OPERATOR Example :

EMP

DEPT

<u>ENAME</u>	<u>DEPTNO</u>
A	20
B	30
C	10
D	60

Left

<u>DNAME</u>	<u>DEPTNO</u>
D1	10
D2	20
D3	30
D4	40

Right

Result Table :

ENAME	DEPTNO	DNAME	DEPTNO
A	20	D2	20
C	10	D1	10
B	30	D3	30
D	60	NULL	NULL

WAQTD names and dname of all the employees even though the employees Don't work in any dept .

**SELECT ENAME,DNAME
FROM EMP,DEPT
WHERE EMP.DEPTNO=DEPT.DEPTNO (+);**

ENAME	DNAME
A	D2
C	D1
B	NULL
D	NULL

Right Outer Join:

"It is used to obtain Un-Matched Records of Right Table Along with Matching Records "

SYNTAX:

ANSI [American National Standard Institute]

SELECT Column_Name
FROM Table_Name1 **RIGHT[OUTER] JOIN** Table_Name2
ON < JOIN_CONDITION> ;

SELECT *
FROM EMP **RIGHT JOIN** DEPT
ON EMP.DEPTNO = DEPT.DEPTNO ;

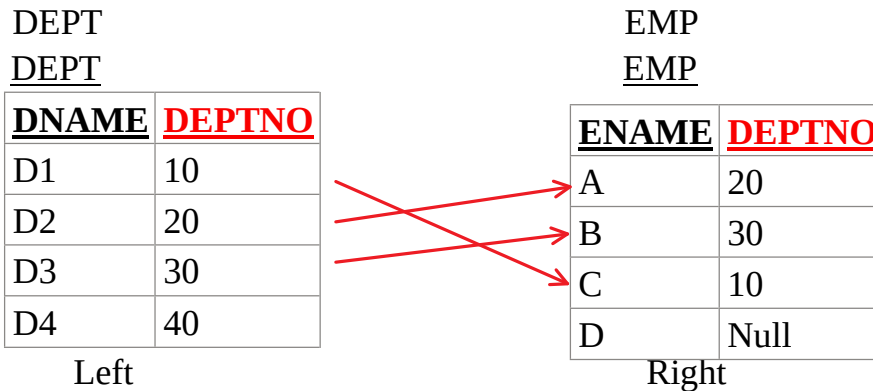
Oracle

SELECT Column_Name
FROM Table_Name1 , Table_Name2

WHERE Table1.Col_Name (+) = Table2.Col_Name ;

SELECT *
FROM EMP , DEPT
WHERE EMP.DEPTNO(+) = DEPT.DEPTNO ;

Example :



Result Table :

<u>DNAME</u>	<u>DEPTNO</u>	<u>ENAME</u>	<u>DEPTNO</u>
D1	10	C	10
D2	20	A	20
D3	30	B	30
NULL	NULL	D	NULL

WAQTD names and dname of all the employees even though the there are no employees in a dept .

SELECT ENAME,DNAME
FROM EMP,DEPT
WHERE EMP.DEPTNO(+)=DEPT.DEPTNO;

Full Outer Join :

"It is used to obtain Un-Matched Records of both Left & Right Table Along with Matching Records ".

SYNTAX

1. **ANSI** [American National Standard Institute]

SELECT Column_Name
FROM Table_Name1 **FULL [OUTER] JOIN** Table_Name2
ON < JOIN_CONDITION> ;

SELECT *
FROM EMP **FULL JOIN** DEPT
ON EMP.DEPTNO = DEPT.DEPTNO ;

Example :

EMP

<u>ENAME</u>	<u>DEPTNO</u>
<u>E</u>	<u>O</u>
A	20
A	20
B	Null
C	10
D	Null

DEPT

<u>DNAME</u>	<u>DEPTNO</u>
D1	10
D1	10
D2	20
D3	30
D4	40

ENAME	DEPTNO	DNAME	DEPTNO
A	20	D2	20
C	10	D1	10
B	NULL	NULL	NULL
D	NULL	NULL	NULL
NULL	NULL	D3	30
NULL	NULL	D4	40

Result Table :

MATCHING RECORDS OF BOTH TABLE

UNMATCHED RECORDS OF LEFT TABLE

UNMATCHED RECORDS OF RIGHT TABLE

WAQTD names and dname of all the employees and depts even though the employees Don't work in any dept and a dept having no employees.

```
SELECT ENAME,DNAME
FROM EMP FULL JOIN DEPT
ON EMP.DEPTNO = DEPT.DEPTNO;
```

SELF JOIN :

"Joining a table by itself is known as Self Join "

Why ? / When ?

"Whenever the data to select is in the same table but present In different records we use self-join ".

SYNTAX:

ANSI [American National Standard Institute]

```
SELECT Column_Name
FROM Table_Name1 JOIN Table_Name2
ON < JOIN_CONDITION > ;
```

```
SELECT *
FROM EMP E1 JOIN EMP E2
ON E1.MGR = E2.EMPNO ;
```

Oracle Oracle

```
SELECT Column_Name
FROM Table_Name1T1 , Table_Name2 T2
WHERE < Join_Condition > ;
```

```
SELECT *
FROM EMP E1 , EMP E2
WHERE E1.MGR = E2.EMPNO;
```

EXAMPLE

EMP E1

EMPNO	ENAME	E1.MGR
1	ALLEN	3
2	SMITH	1
3	MILLER	2

JOIN CONDITON **E1.MGR = E2.EMPNO**

RESULT TABLE;

E1.EMPNO	E1.ENAME	E1.MGR	E2.EMPNO	E2.ENAME	E2.MRG
1	ALLEN	3	3	MILLER	2
2	SMITH	1	1	ALLEN	3
3	MILLER	2	2	SMITH	1

EMP DETAILS - E1

MANAGERS DETAILS -E2

1. WAQTD Ename and Manager's name for all the employees .

```
SELECT E1.ENAME ,E2.ENAME MGR_NAME
FROM EMP E1,EMP E2
WHERE E1.MGR=E2.EMPNO;
```

2. WAQTD Ename , sal along with manager's name and manager's salary for all the employees .

```
SELECT E1.ENAME,E1.SAL, E2.ENAME MGR_NAME ,E2.SAL MGR_SAL
```

```
FROM EMP E1,EMP E2
WHERE E1.MGR=E2.EMPNO ;
```

3. WAQTD Ename , manager's name along with their deptno If employee is working as clerk .

```
SELECT E1.ENAME, E2.ENAME MGR_NAME, E1.DEPTNO ,E2.DEPTNO
MGR_DEPT
FROM EMP E1,EMP E2
WHERE E1.MGR=E2.EMPNO AND E1.JOB IN ' CLERK';
```

NOTE :

TO join 'N' number of tables we need to write 'N-1' number of join conditions

5. NATURAL JOIN:

"It behaves as **INNER JOIN** if there is a relation between the given two tables , else it behaves as **CROSS JOIN**" .

SYNTAX:

ANSI:

```
SELECT Col_Name
FROM Table_Name1 NATURAL JOIN Table_Name2;
```

EXAMPLE

EMP

ENAME	DEPTNO
A	20
B	30
C	10

DEPT

DNAME	DEPTNO
D1	10
D2	20
D3	30

RESULT TABLE:

ENAME	DEPTNO	DNAME
A	20	D2
B	30	D3
C	10	D1

EMP

<u>ENAME</u>	<u>DEPTNO</u>
A	20
B	30
C	10

CUSTOMER

<u>CNAME</u>	<u>CID</u>
X	101
Y	102
Z	103

RESULT TABLE

ENAME	DEPTNO	CNAME	CID
A	20	X	101
A	20	Y	102
A	20	Z	103
B	30	X	101
B	30	Y	102
B	30	Z	103
C	10	X	101
C	10	Y	102
C	10	Z	103

EXAMPLE (NOTE)/ (BLOCK DIAGRAM)

WAQTD ENAME AND DNAME FOR ALL THE EMP.

EMP

DEPT

EMP.DEPTNO =DEPT. DEPTNO

WAQTD ENAME,DNAME AND MANGER'S NAME

E1.MGR=E2.EMPNO

EMP

DEPT

EMP E2

3-1=2

EMP.DEPTNO=DEPT.DEPTNO

SELECT E1.ENAME,DNAME,E2.ENAME

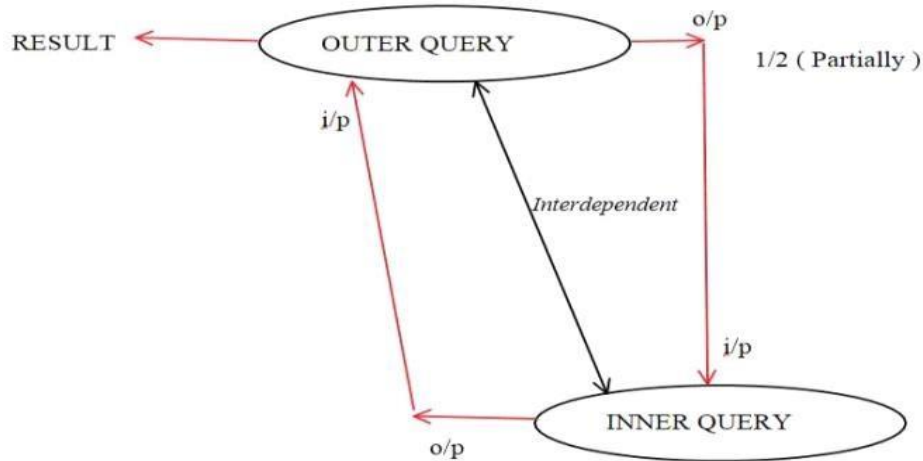
FROM EMP E1,EMP E2,DEPT

WHERE EMP.DEPTNO=DEPT.DEPTNO AND E1.MGR=E2.EMPNO;

CO - RELATED SUB QUERY

" A query written inside another query such that the outer query and the inner query are Dependent on each other , this is known as Co-Related Sub-Query ".

WORKING PRINCIPLE :



Let us consider two queries inner and outer query respectively ,

1. Outer query executes first but partially (50 PER)
2. The partially executed output is given as an input to the inner Query
3. The inner query executes completely and generates an output
4. The output of inner query is fed as an input to the Outer query and Outer Query produces the result .
5. Therefore, we can state that the outer query and the inner query both are INTERDEPENDENT (dependent on each other) .

NOTE :

i. In co-related sub query a Join condition is a must , And must be written only in the Inner Query . ii. Co-Related sub query works with the principles of both SUB QUERY & JOINS .

DIFFERENCE BETWEEN SUB QUERY AND CO RELATED SUB QUERY .

SUB QUERY	CO-RELATED SUB QUERY
Inner query executes first	Inner and outer query partially executes first
Outer query is dependent on inner query	Both are interdependent
Join condition not mandatory	Join condition is mandatory and must be written in inner query
Outer query executes Once	Outer query executes Twice .

Example :

DEPT

<u>DNAME</u>	<u>DNO</u>
D1	10
D2	20
D3	30
D4	40

EMP

<u>ENAME</u>	<u>DNO</u>
A	20
B	10
C	20
D	30

D1 D2 D3 NULL
SELECT DNAME
FROM DEPT
WHERE DNO IN (SELECT DNO
FROM EMP
WHERE EMP.DEPTNO=DEPT.DEPTNO);

10
20
30
40

10
(20,20)
30
NULL

20
10
20
30

10
20
30
40

10=10	TRUE
10=20	F
10=20	F
10=30	F
20=10	F
20=20	T
20=20	T
20=30	F

30=10	F
30=20	F
30=20	F
30=30	T
40=10	F
40=20	F
40=20	F
40=30	F

EX:

1. WAQTD dname in which there are employees working .
SELECT DNAME,DEPTNO FROM
DEPT
WHERE DEPTNO IN (SELECT DEPTNO
FROM EMP
WHERE EMP.DEPTNO=DEPT.DEPTNO);
2. WAQTD dname in which there are no employees working .
SELECT DNAME,DEPTNO FROM
DEPT
WHERE DEPTNO NOT IN (SELECT DEPTNO
FROM EMP
WHERE EMP.DEPTNO=DEPT.DEPTNO);

EXISTS & NOT EXISTS OPERATORS :

1. **EXISTS :** "Exists Op is a Unary Op (One Operand) which can accept one operand towards RHS and that operand has to be co related sub -query

○ *Exists Op returns true if the Sub Query returns Any value other than Null.*

example:

d1,d2,d3

SELECT DNAME

FROM DEPT

WHERE **exists** (SELECT DNO
FROM EMP
WHERE **EMP.DEPTNO=DEPT.DEPTNO**);

10
20
30
NULL

10=10	TRUE
10=20	F
10=20	F
10=30	F
20=10	F
20=20	T
20=20	T
20=30	F

30=10	F
30=20	F
30=20	F
30=30	T
40=10	F
40=20	F
40=20	F
40=30	F

2. **NOT EXISTS :**

"NOT EXISTS operator is a unary op(one operand) which can accept one operand towards RHS and that operand has to a co related sub query"

○ *Not Exists Op returns true if the Sub Query returns NULL .*

To Find MAX & MIN salary :

To find MAXIMUM salary :

```
SELECT distinct SAL
FROM EMP E1
WHERE N = ( SELECT COUNT( DISTINCT SAL )
            FROM EMP E2
            WHERE E1.SAL <= E2.SAL );
```

4TH MAX SAL , 8TH, 10TH

<u>E1.SAL</u>	
1000	SELECT SAL
3000	FROM EMP E1
2000	WHERE (SELECT COUNT(DISTINCT SAL)
	FROM EMP E2
	WHERE E1.SAL < E2.SAL) IN (3,7,9);
	3000
	3000
	2000
	4000
	5000

To find MINIMUM salary :

```
SELECT distinct SAL
FROM EMP E1
WHERE N = ( SELECT COUNT( DISTINCT SAL )
            FROM EMP E2
            WHERE E1.SAL > = E2.SAL ) ;
```